

Laser-induced nonsequential double ionization at and above the recollision-excitation-tunneling threshold

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We perform a rigorous, semianalytic study of the recollision excitation with subsequent tunneling ionization (RESI) mechanism in laser-induced nonsequential double ionization (NSDI), based on the strong-field approximation. We show that the shapes of the electron momentum distributions carry information about the bound state with which the first electron collides, the bound state to which the second electron is excited, and the type of electron-electron interaction. Furthermore, one may define a driving-field intensity threshold for the RESI physical mechanism. At the threshold, the kinetic energy of the first electron, upon return, is just sufficient to excite the second electron. We compute the distributions for helium and argon in the threshold and above-threshold intensity regimes. In the latter case, we relate our findings to existing experiments. The electron momentum distributions encountered are symmetric with respect to all quadrants of the plane spanned by the momentum components parallel to the laser-field polarization, instead of concentrating on only the second and fourth quadrants. The above-mentioned momentum constraints, together with the strong dependence of the distributions on the bound states involved, may be important for singling out the RESI mechanism in actual physical situations and using NSDI in ultrafast imaging.

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I. INTRODUCTION

Electron-electron correlation in strong laser fields has raised considerable interest for over a decade, in particular in the context of laser-induced nonsequential double and multiple ionization [1]. Concrete examples are the early measurements of a “knee” in the double ionization yield as a function of the laser-field intensity, which deviates from the predictions of sequential models in orders of magnitude [2], and the peaks in the electron momentum distributions in nonsequential double ionization (NSDI), as functions of the electron components $p_{n\parallel}$ ($n = 1, 2$) parallel to the laser-field polarization [3]. Such peaks occur at nonvanishing parallel momenta and cannot be explained by a sequential mechanism.

Presently, it is an established fact that NSDI occurs due to the inelastic recollision of an electron with its parent ion [4]. In this recollision, the first electron gives part of the kinetic energy it acquired from the driving field to a second electron, which is then freed.

The simplest type of recollision which can lead to this phenomenon is electron-impact ionization. Thereby, the first electron, upon return, provides the second electron with enough energy so that it is able to overcome the second ionization potential of the target in question and reach the continuum. Both electrons leave simultaneously and lead to distributions peaked at nonvanishing momenta, occupying the first and third quadrant of the plane $p_{1\parallel} p_{2\parallel}$ spanned by the parallel momentum components.

Electron-impact ionization is possibly the most extensively investigated NSDI mechanism [5–13]. This is due to the fact that, until the past few years, it was sufficient to describe key features observed experimentally in the electron momentum distributions. Concrete examples are the peaks near the nonvanishing momenta $p_{1\parallel} = p_{2\parallel} = \pm 2\sqrt{U_p}$, where U_p is the ponderomotive energy, and the recently reported V-shaped structure [11], which is a signature of the long-range character

of the electron-electron interaction [7,8,10,12,13]. Moreover, from the theoretical viewpoint, this mechanism is considerably easier to model compared to other rescattering processes. This is particularly true in the context of semianalytical models, such as the strong-field approximation [5–7,9,13,14].

Recent experimental results, however, reveal other rescattering mechanisms apart from electron-impact ionization. For instance, if the driving-field intensity is below the threshold intensity, the kinetic energy of the recolliding electron is no longer sufficient to make the second electron overcome the ionization potential of the singly ionized core and reach the continuum [15,16]. In this intensity range, the energy transferred to the core is only enough to raise the second electron to an excited bound state, from which it subsequently tunnels. This process is known as recollision excitation with subsequent tunneling ionization (RESI). This mechanism is also important for specific targets, such as argon [17], even if intensities above the threshold are taken, provided the driving pulse is long enough.

For this specific mechanism, the first electron leaves near a crossing of the driving field, while the second electron is freed near a field maximum. The time delay between both electrons suggests that they leave with opposite momenta. This implies that the second and the fourth quadrants of the $p_{1\parallel} p_{2\parallel}$ plane are expected to be populated [18]. A population in such regions of the parallel momentum plane has also been observed for NSDI of diatomic molecules, especially if the molecules in question are aligned perpendicular to the laser-field polarization [19–22]. Furthermore, it has been recently shown that electron-impact ionization alone would not be very useful for retrieving the molecular structure in the experimentally relevant intensity range [23]. All these examples suggest that the RESI mechanism is becoming increasingly important, in view of the driving-field intensities and targets employed.

This mechanism, however, is considerably less understood than electron-impact ionization. In fact, apart from early results

in which only the RESI yield has been calculated within a semianalytical framework [14], most existing results are the outcome of computations, either classical [18–20,24,25] or quantum mechanical [22], for which the different rescattering mechanisms cannot be easily disentangled. In fact, in many of the above-cited studies, RESI is masked by electron-impact ionization, or multiple collisions. In particular, electron-impact ionization populates the first and third quadrants of the parallel momentum plane and may lead to the conclusion that RESI does not lead to probabilities in such regions.

Apart from that, the RESI mechanism is very difficult to model within a classical framework. For that reason, in many classical studies one tries to mimic this process by taking electron-impact ionization with a time delay between the rescattering of the first electron and the ionization of the second electron [18,20,21,24]. However, this is a different physical mechanism from RESI and leads to different momentum constraints.

In our previous work [26], we have studied the RESI physical mechanism within the strong-field approximation. In this framework, the transition amplitude is written as a multiple integral, with a time-dependent action and slowly varying prefactors. This approach has the particular advantage of providing a clear space-time picture of the process in question, in terms of electron trajectories, while retaining features such as quantum-interference effects [27].

We have shown that the RESI mechanism may be understood as the combination of two processes. For the first electron, a behavior similar to rescattered above-threshold ionization is present. The main difference lies in the fact that part of the kinetic energy of the first electron is given to the parent ion. This leads to a maximal kinetic energy slightly smaller than the high-order above-threshold ionization (ATI) cutoff $10U_p$. The second electron, which is tunnel ionized at a subsequent time, behaves in the same way as in direct above-threshold ionization. Hence, its maximal kinetic energy is given by the direct ATI cutoff (i.e., $2U_p$). This allows one to determine kinematic constraints for the RESI mechanism, which lead to distributions that equally occupy the four quadrants of the $p_{1\parallel}p_{2\parallel}$ plane. For an early derivation of somewhat different kinematic constraints for this mechanism, see, for example, Ref. [28].

In the above-stated investigations, however, we assumed the prefactors in the strong-field approximation amplitude to be constant. Nonetheless, in a more realistic scenario, one should take into account that such prefactors cause a bias in momentum space. This may lead to electron momentum distributions concentrated on the second and fourth quadrants of the $p_{1\parallel}p_{2\parallel}$ plane. Apart from that, a legitimate question is whether the state in which the second electron is initially bound, or to which it is excited, affects the shapes of the RESI electron momentum distributions in a dramatic fashion. If this is indeed the case, the electron momentum distributions carry an imprint of such states. Hence, nonsequential double ionization may be employed for resolving excitation in the attosecond domain.

In this work, we address these particular questions within a rigorous, semianalytic framework. We perform a comprehensive analysis of the role of the states in which the second electron is initially bound and to which it is excited, and the type of interaction by which the excitation takes place. We also iden-

tify a threshold driving-field intensity for the RESI mechanism, which, physically, corresponds to the situation in which the first electron has just enough energy to excite the second. This intensity is below the threshold for electron-impact ionization.

This work is organized as follows. In Sec. II, we briefly recall the strong-field approximation (SFA) transition amplitude for the RESI mechanism, which was derived in Ref. [26]. We start from the general expressions (Sec. II A), which are solved by saddle-point methods. The pertaining saddle-point equations are derived in Sec. II B. Subsequently, in Sec. II C we provide the specific prefactors employed for the hydrogenic systems to be investigated in this work. In Sec. III, we employ this approach to compute electron momentum distributions for helium and argon. For the latter species, an explicit comparison with the results in Ref. [15] is performed. Finally, in Sec. IV we state the main conclusions of the paper.

II. TRANSITION AMPLITUDE

A. General expressions

The transition amplitude describing the RESI mechanism, within the SFA,¹ can be written as (for details on the derivation see Ref. [26])

$$M(\mathbf{p}_1, \mathbf{p}_2) = \int_{-\infty}^{\infty} dt \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' \int d^3k \times V_{\mathbf{p}_2e} V_{\mathbf{p}_1e, \mathbf{k}g} V_{\mathbf{k}g} e^{iS(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'')}, \quad (1)$$

with the action

$$\begin{aligned} S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') &= - \int_t^{\infty} \frac{[\mathbf{p}_2 + \mathbf{A}(\tau)]^2}{2} d\tau - \int_{t'}^{\infty} \frac{[\mathbf{p}_1 + \mathbf{A}(\tau)]^2}{2} d\tau \\ &\quad - \int_{t''}^{t'} \frac{[\mathbf{k} + \mathbf{A}(\tau)]^2}{2} d\tau + E_{1g}t'' + E_{2g}t' + E_{2e}(t - t') \end{aligned} \quad (2)$$

and the prefactors

$$\begin{aligned} V_{\mathbf{k}g} &= \langle \tilde{\mathbf{k}}(t'') | V | \psi_g^{(1)} \rangle \\ &= \frac{1}{(2\pi)^{3/2}} \int d^3r_1 V(r_1) \exp[-i\tilde{\mathbf{k}}(t'') \cdot \mathbf{r}_1] \psi_g^{(1)}(\mathbf{r}_1), \end{aligned} \quad (3)$$

$$\begin{aligned} V_{\mathbf{p}_1e, \mathbf{k}g} &= \langle \tilde{\mathbf{p}}_1(t'), \psi_e^{(2)} | V_{12} | \tilde{\mathbf{k}}(t'), \psi_g^{(2)} \rangle \\ &= \frac{1}{(2\pi)^3} \iint d^3r_2 d^3r_1 \exp[-i(\mathbf{p}_1 - \mathbf{k}) \cdot \mathbf{r}_1] \\ &\quad \times V_{12}(\mathbf{r}_1, \mathbf{r}_2) [\psi_e^{(2)}(\mathbf{r}_2)]^* \psi_g^{(2)}(\mathbf{r}_2) \end{aligned} \quad (4)$$

and

$$\begin{aligned} V_{\mathbf{p}_2e} &= \langle \tilde{\mathbf{p}}_2(t) | V_{\text{ion}} | \psi_e^{(2)} \rangle \\ &= \frac{1}{(2\pi)^{3/2}} \int d^3r_2 V_{\text{ion}}(r_2) \exp[-i\tilde{\mathbf{p}}_2(t) \cdot \mathbf{r}_2] \psi_e^{(2)}(\mathbf{r}_2). \end{aligned} \quad (5)$$

¹The SFA consists of the neglect of the residual ionic potentials when the electrons are in the continuum and of the external laser field when the electrons are bound. One should note, however, that the SFA does not *per se* exclude electron-electron correlation or excitation, and it can also be applied to multielectron systems.

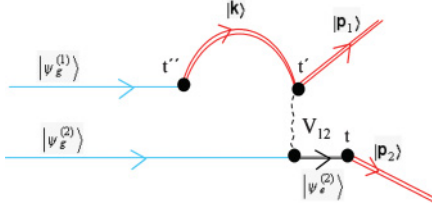


FIG. 1. (Color online) Feynman diagram of the process described by the transition amplitude (1), in which an electron, initially in a bound state $|\psi_g^{(1)}\rangle$, is released by tunneling ionization into the Volkov state in a time t'' , returns at a time t' and excites a second electron from $|\psi_g^{(2)}\rangle$ to the state $|\psi_e^{(2)}\rangle$, from which it subsequently tunnels at a later time t , reaching a Volkov state. The electron-electron interaction is indicated in the figure by V_{12} , the initial bound states by the dark blue lines, the excited bound state of the second electron by the thick black line, and the Volkov states by the double (red) lines.

Equation (1) describes the physical process in which an electron, initially in a bound state $|\psi_g^{(1)}\rangle$, is released by tunneling ionization at a time t'' into a Volkov state $|\tilde{\mathbf{k}}(t'')\rangle$. Subsequently, this electron propagates in the continuum from t'' to a later time t' . At this time, it is driven back by the field and rescatters with its parent ion. In this collision, it excites the second electron, which is bound at $|\psi_g^{(2)}\rangle$, to the state $|\psi_e^{(2)}\rangle$ through the interaction V_{12} . Finally, the second electron, which is in a bound excited state $|\psi_e^{(2)}\rangle$, is released by tunneling ionization at a time t into a Volkov state $|\tilde{\mathbf{p}}_2(t)\rangle$. The final electron momenta are described by $\mathbf{p}_n$ ($n = 1, 2$). In the above-stated equations, E_{ng} ($n = 1, 2$) give the ionization potentials of the ground state, E_{ne} ($n = 1, 2$) give the ionization potentials of the excited state, and $V(r_1)$ and $V_{\text{ion}}(r_2)$ correspond to the atomic binding potential of the system as seen by the first and second electron, respectively. The Feynman diagram corresponding to this process is depicted in Fig. 1.

The form factors (3) and (5) contain all the information about the binding potential of the first and second electron, respectively. The form factor (4) contains all the information about the interaction of the first electron with the singly ionized atom, which is an inelastic interaction. If the electron interaction is only dependent on the difference between both electron coordinates, that is, if $V_{12}(\mathbf{r}_1, \mathbf{r}_2) = V_{12}(\mathbf{r}_1 - \mathbf{r}_2)$, Eq. (4) may be rewritten as

$$V_{\mathbf{p}_1 e, \mathbf{k} g} = \frac{V_{12}(\mathbf{p}_1 - \mathbf{k})}{(2\pi)^{3/2}} \int d^3 r_2 e^{-i(\mathbf{p}_1 - \mathbf{k}) \cdot \mathbf{r}_2} [\psi_e^{(2)}(\mathbf{r}_2)]^* \psi_g^{(2)}(\mathbf{r}_2), \quad (6)$$

with

$$V_{12}(\mathbf{p}_1 - \mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int d^3 r \exp[-i(\mathbf{p}_1 - \mathbf{k}) \cdot \mathbf{r}] V_{12}(\mathbf{r}) \quad (7)$$

and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. One should note that, in our framework, electron-electron correlation is embedded in the excitation process, and the first electron interacts with the second via the prefactor $V_{\mathbf{p}_1 e, \mathbf{k} g}$.

Clearly, $V_{\mathbf{k} g}$ and $V_{\mathbf{p}_2 e}$ are gauge dependent. In fact, in the length gauge $\tilde{\mathbf{p}}_n(\tau) = \mathbf{p}_n + \mathbf{A}(\tau)$ and $\tilde{\mathbf{k}}(\tau) = \mathbf{k} + \mathbf{A}(\tau)$ ($\tau = t', t''$), while in the velocity gauge $\tilde{\mathbf{p}}_n(\tau) = \mathbf{p}_n$ and $\tilde{\mathbf{k}}(\tau) = \mathbf{k}$. This is a direct consequence of the fact that the

gauge transformation $\chi_{l \rightarrow v} = \exp[-i\mathbf{A}(\tau) \cdot \mathbf{r}]$ from the length to the velocity gauge causes a translation $\mathbf{p} \rightarrow \mathbf{p} - \mathbf{A}(\tau)$ in momentum space. Due to the fact that these shifts cancel out in Eq. (6), $V_{\mathbf{p}_1 e, \mathbf{k} g}$ remains the same in the length and velocity gauges.

B. Saddle-point analysis

The multiple integral in Eq. (1) is solved using saddle-point methods (for details see Ref. [29]). For that purpose, we must find the coordinates $(t_s, t'_s, t''_s, \mathbf{k}_s)$ for which $S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'')$ is stationary, that is, for which the conditions $\partial_t S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = \partial_{t'} S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = \partial_{t''} S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = 0$ and $\partial_{\mathbf{k}} S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = \mathbf{0}$ are satisfied. This leads to the equations

$$[\mathbf{k} + \mathbf{A}(t'')]^2 = -2E_{1g}, \quad (8)$$

$$\mathbf{k} = -\frac{1}{t' - t''} \int_{t''}^{t'} d\tau \mathbf{A}(\tau), \quad (9)$$

$$[\mathbf{p}_1 + \mathbf{A}(t')]^2 = [\mathbf{k} + \mathbf{A}(t')]^2 - 2(E_{2g} - E_{2e}), \quad (10)$$

and

$$[\mathbf{p}_2 + \mathbf{A}(t)]^2 = -2E_{2e}. \quad (11)$$

Equation (8) gives the conservation of energy at time t'' , which physically corresponds to the tunneling of the first electron. Since tunneling has no classical counterpart, this equation possesses no real solution. Equation (9) constrains the intermediate momentum $\mathbf{k}$ of the first electron so that it can return to the parent ion. Equation (10) expresses the fact that the first electron returns to its parent ion at time t' and rescatters inelastically with it, giving part of its kinetic energy $E_{\text{ret}}(t') = [\mathbf{k} + \mathbf{A}(t')]^2/2$ to the core. Under this interaction, the second electron is excited from a state with energy E_{2g} to a state with energy E_{2e} . The first electron leaves immediately and reaches the detector with momentum $\mathbf{p}_1$. Finally, Eq. (11) describes the fact that the second electron tunnels at a later time t from an excited state of energy E_{2e} .

1. Momentum constraints

The saddle-point Eqs. (10) and (11) provide useful information on the momentum-space regions populated by the RESI mechanism and on the shapes of the electron momentum distributions. For instance, Eq. (11) is identical to that describing tunnel ionization for direct above-threshold ionization. This implies that the maximal kinetic energy of the second electron at the detector, if the field can be approximated by a monochromatic wave, is roughly given by $2U_p$.

Furthermore, the electron is leaving the excited state with the largest probability when the electric field $E(t) = -dA(t)/dt$ is at its maximum. If the time dependence of the laser field is such that $A(t)$ vanishes when $E(t)$ is at its peak (for instance, monochromatic fields), then

$$-2\sqrt{U_p} \leq p_2 \leq 2\sqrt{U_p}. \quad (12)$$

If, to first approximation, we neglect the momentum components perpendicular to the laser-field polarization, one can see that the momentum of the second electron, in the parallel momentum plane, is expected to be centered around vanishing momentum $p_{2\parallel}$ and limited by the bounds $p_{2\parallel} = \pm 2\sqrt{U_p}$.

One should note that, due to the fact that Eq. (11) describes a tunneling process, there is no classically allowed region for the second electron. A nonvanishing perpendicular momentum effectively leads to an increase in the potential barrier and a suppression in the yield. In fact, Eq. (11) can also be written as

$$[p_{2\parallel} + A(t)]^2 = -2\tilde{E}_{2e}, \quad (13)$$

where $\tilde{E}_{2e} = E_{2e} + \mathbf{p}_{2\perp}^2$ is an effective ionization potential. Hence, the barrier effectively widens for nonvanishing transverse momenta, leading to an overall decrease in the tunneling probability (see Ref. [26] for details).

The saddle-point Eq. (10), on the other hand, yields information about the momentum of the first electron. According to this equation,

$$-A(t') - \sqrt{2E_{\text{diff}}} \leq p_{1\parallel} \leq -A(t') + \sqrt{2E_{\text{diff}}}, \quad (14)$$

where $E_{\text{diff}} = E_{\text{kin}}(t', t'') - (E_{2g} - E_{2e}) - \mathbf{p}_{1\perp}^2/2$ and $E_{\text{kin}}(t', t'')$ denotes the kinetic energy of the first electron upon return. For a monochromatic field, the electron returns most probably near a crossing of the laser field; one may use the approximation $A(t) \simeq 2\sqrt{U_p}$ in the above-stated equation. In this case, we also know that the kinetic energy $E_{\text{kin}}(t', t'') \leq 3.17U_p$. Hence, $E_{\text{diff}}^{(\text{max})} \leq 3.17U_p - (E_{2g} - E_{2e}) - \mathbf{p}_{1\perp}^2/2$ and

$$-2\sqrt{U_p} - \sqrt{2E_{\text{diff}}^{(\text{max})}} \leq p_{1\parallel} \leq -2\sqrt{U_p} + \sqrt{2E_{\text{diff}}^{(\text{max})}}. \quad (15)$$

Equation (15) allows one to delimit a region in momentum space for $p_{1\parallel}$ centered around $-2\sqrt{U_p}$ and bounded by $2E_{\text{diff}}^{(\text{max})}$. In contrast to the previous case, there may be a classically allowed region for the momentum of the first electron if the parameters inside the square root are positive, that is, if $3.17U_p \geq (E_{2g} - E_{2e}) + \mathbf{p}_{1\perp}^2/2$. For increasing perpendicular momentum and/or bound-state energy difference, this region becomes more and more localized around $-2\sqrt{U_p}$ until it collapses. Therefore, it is also possible to distinguish a threshold and an above-threshold behavior in the context of recollision-excitation tunneling. One should note, however, that intensities below the recollision-excitation threshold $(E_{2g} - E_{2e}) = 3.17U_p$ do not make physical sense, as the energy of the returning electron would not be sufficient to promote the bound electron to an excited state. If $(E_{2g} - E_{2e}) \ll 3.17U_p$, the well-known cutoff of $10U_p$ for rescattered above-threshold ionization is recovered.²

In view of the above-mentioned constraints, the expected maxima of the electron momentum distribution are located at the most probable momenta $(p_{1\parallel}, p_{2\parallel}) = (\pm 2\sqrt{U_p}, 0)$ and, after symmetrizing with respect to the exchange $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$, at $(p_{1\parallel}, p_{2\parallel}) = (0, \pm 2\sqrt{U_p})$. This implies that, if the field can be approximately described by a monochromatic wave, the outcome of our model should be distributions in the $p_{1\parallel} p_{2\parallel}$ plane, which are symmetric upon $\mathbf{p}_n \rightarrow -\mathbf{p}_n$ ($n = 1, 2$) and

upon $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$, and which equally occupy the four quadrants of the parallel momentum plane.

2. Bound-state singularity

Finally, due to the saddle-point Eqs. (8) and (11), for exponentially decaying bound states the prefactors (3) and (5) exhibit singularities in the length-gauge formulation of the strong-field approximation. This is due to the fact that these prefactors are inversely proportional to $([\mathbf{k} + \mathbf{A}(t'')]^2 + 2E_{1g})^n$ and $([\mathbf{p}_2 + \mathbf{A}(t)]^2 + 2E_{2e})^m$, where n and m are integers. For the problem addressed in this specific work, however, only the prefactor $V_{\mathbf{p}_{2e}}$ influences the shape of the electron momentum distributions. This is a consequence of the fact that it gives the final momentum $\mathbf{p}_2$ of the second electron at the detector. The prefactor $V_{\mathbf{k}_g}$, in contrast, determines the intermediate momentum $\mathbf{k}$, which changes to $\mathbf{p}_1$ after recollision. Hence, it only affects the distributions quantitatively. Therefore, to first approximation, one can consider Eq. (3) to be constant. A similar problem for the electron-impact ionization mechanism in NDSI has been discussed in detail in Ref. [7].

To overcome the singularity in $V_{\mathbf{p}_{2e}}$, one needs to embed this prefactor into the action, which now reads

$$\tilde{S}(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') - i \ln V_{\mathbf{p}_{2e}}. \quad (16)$$

This leads to modifications in the saddle-point equation $\partial_t \tilde{S}(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'')$, which is now given by

$$[\mathbf{p}_2 + \mathbf{A}(t)]^2 = -2E_{2e} + i \partial_t \ln V_{\mathbf{p}_{2e}}. \quad (17)$$

The main consequence of such a modification is that the drift velocity of the second electron is no longer pure imaginary. This leads to a splitting in the ionization time t for each orbit, as compared to the nonmodified case. Depending on the velocity in question, the barrier the electron must tunnel through in order to reach the continuum will either widen or narrow. This means that, with regard to the nonmodified action, $\text{Im}[t]$ will either increase or decrease.

C. Prefactors

In this work, we are particularly interested in exponentially decaying hydrogenic bound states. This means that, in general, the bound-state wave function reads

$$\psi^{(\alpha)}(\mathbf{r}_\alpha) = R_{nl}(r_\alpha) Y_l^m(\theta_\alpha, \varphi_\alpha), \quad (18)$$

where n , l , and m denote the principal, orbital, and magnetic quantum numbers; the index α refers to the electron in question; and the angular coordinates are given by θ_α and φ_α . In this case, the binding potentials $V(r_1)$ and $V_{\text{ion}}(r_2)$ are given by

$$V_\alpha(r_\alpha) = -\frac{Z_{\text{eff}}}{r_\alpha}, \quad (19)$$

where V_α yields either V or V_{ion} , and Z_{eff} corresponds to the effective electronic charge. The general expressions for the prefactors in this work are provided in the Appendix A.

Below, we state the specific prefactors to be employed in Sec. III, for helium and argon. In the former case, upon collision, the second electron may be excited from the $1s$ state to either the $2s$ or the $2p$ state, while in the latter species it may undergo a transition from the $3p$ state to the $4s$ or the $4p$ state.

²Different bounds have been provided in our previous publication [26]. Such bounds were derived based on the above-threshold ionization kinetic energy values. Hence, they are only applicable when $E_{\text{kin}}(t', t'') \gg E_{2g} - E_{2e}$, while the present bounds are valid throughout.

One should note that the prefactor $V_{\mathbf{p}_2e}$ is gauge dependent. In the length gauge, $\tilde{\mathbf{p}}_2(t) = \mathbf{p}_2 + \mathbf{A}(t)$ and $\tilde{p}_2(t)_{\parallel} = p_{2\parallel} + A(t)$ while, in the velocity gauge, $\tilde{\mathbf{p}}_2(t) = \mathbf{p}_2$ and $\tilde{p}_2(t)_{\parallel} = p_{2\parallel}$. The prefactor $V_{\mathbf{p}_1e, \mathbf{k}g}$, on the other hand, is gauge invariant.

1. Excitation $1s \rightarrow 2s$

Let us first consider the simplest case, in which the second electron is excited to $2s$. This gives the prefactors

$$V_{\mathbf{p}_2e}^{(2s)} \sim \frac{[\tilde{p}_2(t)]^2 - 2E_{2e}}{\{[\tilde{p}_2(t)]^2 + 2E_{2e}\}^2} \quad (20)$$

and

$$V_{\mathbf{p}_1e, \mathbf{k}g}^{(1s \rightarrow 2s)} \sim V_{12}(\mathbf{p}_1 - \mathbf{k}) \frac{\eta_1(\kappa^2, E_{2g}, E_{2e})}{[\kappa^2 + \zeta^2(E_{2g}, E_{2e})]^3}, \quad (21)$$

with

$$\eta_1(\kappa^2, E_{2g}, E_{2e}) = \kappa^2(\sqrt{2E_{2g}} + 2\sqrt{2E_{2e}}) + (2E_{2g})^{3/2} - 2(2E_{2e})^{3/2} - 6E_{2e}\sqrt{2E_{2g}} \quad (22)$$

and

$$\zeta(E_{2g}, E_{2e}) = \sqrt{2E_{2e}} + \sqrt{2E_{2g}}. \quad (23)$$

The above-stated equations can also be written in terms of the momentum components parallel and perpendicular to the laser-field polarization, denoted by $p_{\alpha\parallel}$ and $p_{\alpha\perp}$ ($\alpha = 1, 2$), respectively. In this case,

$$V_{\mathbf{p}_2e}^{(2s)} \sim \frac{[\tilde{p}_2(t)_{\parallel}]^2 + \mathbf{p}_{2\perp}^2 - 2E_{2e}}{\{[\tilde{p}_2(t)_{\parallel}]^2 + \mathbf{p}_{2\perp}^2 + 2E_{2e}\}^2}, \quad (24)$$

$$V_{\mathbf{p}_1e, \mathbf{k}g}^{(1s \rightarrow 2s)} \sim V_{12}(\mathbf{p}_1 - \mathbf{k}) \frac{\eta_1[(k - p_{1\parallel})^2 + \mathbf{p}_{1\perp}^2, E_{2g}, E_{2e}]}{[(k - p_{1\parallel})^2 + \mathbf{p}_{1\perp}^2 + \zeta^2(E_{2g}, E_{2e})]^3}. \quad (25)$$

2. Excitation $1s \rightarrow 2p$

If, on the other hand, the second electron is excited to $2p$, one must consider three degenerate states, corresponding to the magnetic quantum numbers $m = \pm 1, 0$. This yields

$$V_{\mathbf{p}_2e}^{(2p)} \sim \frac{\sqrt{[\tilde{p}_2(t)]^2}}{(2E_{2e} + [\tilde{p}_2(t)]^2)^2} [Y_1^m(\theta_{\tilde{p}_2}, \varphi_{\tilde{p}_2})]^* \quad (26)$$

and

$$V_{\mathbf{p}_1e, \mathbf{k}g}^{(1s \rightarrow 2p)} \sim V_{12}(\mathbf{p}_1 - \mathbf{k}) \eta_2(\kappa^2, E_{2g}, E_{2e}) [Y_1^m(\theta_{\kappa}, \varphi_{\kappa})]^*, \quad (27)$$

with

$$\eta_2(\kappa^2, E_{2g}, E_{2e}) = \frac{\zeta(E_{2g}, E_{2e})\sqrt{\kappa^2}}{[\zeta^2(E_{2g}, E_{2e}) + \kappa^2]^3}. \quad (28)$$

Since the electron may be excited to any of the $2p$ states, we consider the coherent superposition

$$|\psi_{2p}^{(2)}\rangle = \frac{1}{\sqrt{3}}(|\psi_{2p_x}^{(2)}\rangle + |\psi_{2p_y}^{(2)}\rangle + |\psi_{2p_z}^{(2)}\rangle), \quad (29)$$

where $\langle \mathbf{r}_2 | \psi_{2p_j}^{(2)} \rangle = \psi_{2p_j}^{(2)}(\mathbf{r}_2)$, with $j = x, y, z$. This implies that

$$V_{\mathbf{p}_2e}^{(2p)} \sim \frac{\sqrt{[\tilde{p}_2(t)]^2}}{\{2E_{2e} + [\tilde{p}_2(t)]^2\}^2} \beta(\tilde{\mathbf{p}}_2(t)) \quad (30)$$

and

$$V_{\mathbf{p}_1e, \mathbf{k}g}^{(1s \rightarrow 2p)} \sim V_{12}(\mathbf{p}_1 - \mathbf{k}) \eta_2(\kappa^2, E_{2g}, E_{2e}) \beta(\kappa), \quad (31)$$

where the angular dependency is given by

$$\beta(\mathbf{q}) = (\sin \theta_q \cos \varphi_q + \sin \theta_q \sin \varphi_q + \cos \theta_q). \quad (32)$$

Thereby, we employed the usual relations between spherical polar coordinates and the spherical harmonics.

One may write the above-stated expressions in terms of the electron momentum components parallel and perpendicular to the laser-field polarization. In this case, Eq. (30) reads

$$V_{\mathbf{p}_2e}^{(2p)} \sim \frac{\sqrt{[\tilde{p}_2(t)_{\parallel}]^2 + \mathbf{p}_{2\perp}^2}}{\{2E_{2e} + [\tilde{p}_2(t)_{\parallel}]^2 + \mathbf{p}_{2\perp}^2\}^2} \beta(\tilde{\mathbf{p}}_2(t)). \quad (33)$$

In $\beta(\tilde{\mathbf{p}}_2(t))$, the angles $\theta_{\tilde{p}_2}$ and $\varphi_{\tilde{p}_2}$ are given by

$$\theta_{\tilde{p}_2} = \arccos\{\tilde{p}_2(t)_{\parallel} / \sqrt{[\tilde{p}_2(t)_{\parallel}]^2 + \mathbf{p}_{2\perp}^2}\} \quad (34)$$

and $\varphi_{\tilde{p}_2} = \arccos[\tilde{p}_2(t)_x / \tilde{p}_2(t)_{\perp}]$, respectively, where the subscript x denotes the x direction in the perpendicular momentum plane. In Eq. (31), $\kappa^2 = (k - p_{1\parallel})^2 + \mathbf{p}_{1\perp}^2$ and the angles θ_{κ} and φ_{κ} read

$$\theta_{\kappa} = \arccos[(k - p_{1\parallel}) / \sqrt{(k - p_{1\parallel})^2 + \mathbf{p}_{1\perp}^2}] \quad (35)$$

and $\varphi_{\kappa} = \arccos[p_{1x} / p_{1\perp}]$, respectively. This angular dependence is washed out when the transverse momentum components are integrated over (see Sec. III).

3. Excitations $3p \rightarrow 4s$ and $3p \rightarrow 4p$

Finally, we assume that the second electron, initially in $3p$, is excited either to the $4s$ or to the $4p$ state. Similarly to the procedure adopted in the previous section, we consider a coherent superposition of the $3p_x$, $3p_y$, and $3p_z$ states for the initial state of the electron; that is,

$$|\psi_{3p}^{(2)}\rangle = \frac{1}{\sqrt{3}}[|\psi_{3p_x}^{(2)}\rangle + |\psi_{3p_y}^{(2)}\rangle + |\psi_{3p_z}^{(2)}\rangle]. \quad (36)$$

If the electron is excited to the $4s$ state, the excitation prefactor $V_{\mathbf{p}_1e, \mathbf{k}g}^{(3p \rightarrow 4s)}$ will exhibit an angular dependence given by $\beta(\kappa)$, and the tunneling prefactor $V_{\mathbf{p}_2e}^{(4s)}$ will not depend on the angular variables. Both prefactors also have a radial dependence on $\tilde{p}_2(t)$ or κ . If, however, the electron is excited to the $4p$ state, one must take the final state as

$$|\psi_{4p}^{(2)}\rangle = \frac{1}{\sqrt{3}}[|\psi_{4p_x}^{(2)}\rangle + |\psi_{4p_y}^{(2)}\rangle + |\psi_{4p_z}^{(2)}\rangle], \quad (37)$$

that is, as a coherent superposition of $4p_x$, $4p_y$, and $4p_z$. In this case, the angular dependence of $V_{\mathbf{p}_2e}$ is embedded in $\beta(\tilde{\mathbf{p}}_2(t))$. The angular dependence of $V_{\mathbf{p}_1e, \mathbf{k}g}^{(3p \rightarrow 4p)}$ is more complex and involves the sum of the orbital angular momenta of the two electronic bound states involved. Due to the higher quantum numbers involved, the prefactors are messier than those in the previous sections and are not written out explicitly. They can, however, be obtained from the general expressions in Appendix A.

III. ELECTRON MOMENTUM DISTRIBUTIONS

In this section, we compute electron momentum distributions as functions of the momentum components $(p_{1\parallel}, p_{2\parallel})$ parallel to the laser-field polarization. We approximate the external laser field by a monochromatic wave; that is,

$$\mathbf{E}(t) = \varepsilon_0 \sin \omega t \mathbf{e}_x. \quad (38)$$

This is a reasonable approximation for pulses whose duration is of the order of 10 cycles or longer (see, e.g., Ref. [9] for a more detailed discussion). In this case, the electron momentum distributions, when integrated over the transverse momentum components, read

$$F(p_{1\parallel}, p_{2\parallel}) = \iint d^2 p_{1\perp} d^2 p_{2\perp} |M_R(\mathbf{p}_1, \mathbf{p}_2) + M_L(\mathbf{p}_1, \mathbf{p}_2) + \mathbf{p}_1 \leftrightarrow \mathbf{p}_2|^2, \quad (39)$$

where $M_R(\mathbf{p}_1, \mathbf{p}_2)$ is given by Eq. (1) and $d^2 p_{n\perp} = p_{n\perp} dp_{n\perp} d\varphi_{p_n}$. The amplitudes $M_R(\mathbf{p}_1, \mathbf{p}_2)$ and $M_L(\mathbf{p}_1, \mathbf{p}_2)$ are related to the right and left peaks in the electron momentum distributions, respectively. If $\mathbf{A}(t \pm T/2) = -\mathbf{A}(t)$, where $T = 2\pi/\omega$ denotes a field cycle, the actions corresponding to the transition amplitudes M_L and M_R are given by $S_R(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'')$ and $S_L(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'') = S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t \pm T/2, t' \pm T/2, t'' \pm T/2)$. They also obey the symmetry $|S(\mathbf{p}_1, \mathbf{p}_2, \mathbf{k}, t, t', t'')| = |S(-\mathbf{p}_1, -\mathbf{p}_2, -\mathbf{k}, t \pm T/2, t' \pm T/2, t'' \pm T/2)|$. The distributions have also been symmetrized with respect to the exchange $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$. To a good approximation, the quantum-interference terms $[M_\nu(\mathbf{p}_1, \mathbf{p}_2)]^* M_\mu(\mathbf{p}_1, \mathbf{p}_2)$, $\nu \neq \mu$, get washed out upon transverse-momentum integration, so it is sufficient to add the above-stated amplitudes incoherently. In Appendix B, we provide details on why the interference terms can be neglected.

Here, we integrate the transition amplitude over the azimuthal angles φ_{p_n} and, unless otherwise stated, we consider $V_{\mathbf{k}g}$ as constant. We now briefly discuss how the prefactors V_{p_2e} and $V_{p_1e, \mathbf{k}g}$ behave with regard to the integration over φ_{p_n} . Obviously, if the second electron is excited from an s state to an s state, the prefactors V_{p_2e} and $V_{p_1e, \mathbf{k}g}$ do not depend on this parameter. However, if a transition from or to a p state is considered, there will be an angular dependence in such prefactors.

For instance, tunneling ionization from a p state would lead to the argument $\beta(\hat{\mathbf{p}}_2(t))$ in Eq. (30). If excitation from an s state to a p state or vice versa takes place, the angular dependence of the prefactor $V_{p_1e, \mathbf{k}g}$ is given by $\beta(\kappa)$ in Eq. (31). When integrated over the azimuthal angles φ_{p_n} , $|\beta(\kappa)|^2$ and $|\beta(\mathbf{p}_2)|^2$ yield 2π , so the angular dependence of these prefactors can be neglected. If, however, states of higher orbital quantum numbers are involved, or if the initial and excited bound states of the second electron are p states, this dependence is more complex.

The simplest scenario is if all prefactors are nonsingular, such as in the velocity-gauge formulation of the SFA. In this case, they contribute to the electron momentum distributions as $|V_{p_2e}|^2$ and $|V_{p_1e, \mathbf{k}g}|^2$. In the length-gauge SFA, however, V_{p_2e} exhibits a singularity and, therefore, must be incorporated in the action according to Eq. (17). This singularity, however, is present only in the radial part of this prefactor. Therefore,

the angular parts are still slowly varying and may be treated as above. The radial part of the prefactor, however, must be incorporated in the action.

A. Intensity dependence

We commence by having a closer look at how the momentum-space constraints affect the electron momentum distributions for different driving-field intensities. For that purpose, we assume that the prefactors $V_{p_1e, \mathbf{k}g}$ and V_{p_2e} are constant, and we vary the laser-field intensity. For the lowest intensity, the kinetic energy of the returning electron is just enough to promote the second electron to an excited state; that is, we are considering the RESI threshold $E_{2g} - E_{2e} \simeq 3.17U_p$. This intensity, however, is below the electron-impact ionization threshold (i.e., $E_{2g} > 3.17U_p$). The intermediate intensity has been chosen such that $E_{2g} - E_{2e} < 3.17U_p$ (i.e., above the threshold for recollision excitation). Nevertheless, this intensity is not sufficient to make the second electron overcome the ionization potential and be freed by electron-impact ionization. Finally, the highest driving-field intensity considered in this section is far above the recollision-excitation threshold and slightly above the electron-impact ionization threshold. This implies that rescattering is classically allowed for both physical mechanisms. The computations in this section have been performed for helium, and the pertaining results are presented in Fig. 2.

As an overall feature, the distributions exhibit four peaks at $(p_{j\parallel}, p_{v\parallel}) = (\pm 2\sqrt{U_p}, 0)$, with $j, v = 1, 2$ and $j \neq v$. These peaks agree well with the constraints discussed in the previous sections. This holds even if the driving-field intensity is just enough to excite the second electron and the only allowed momenta are $\pm 2\sqrt{U_p}$ [Fig. 2(a)]. Physically, this means that the first electron will reach its parent ion most probably at a crossing of the driving field, reaching the detector with the most probable momenta of $\pm 2\sqrt{U_p}$, whereas the second electron will reach it with vanishing momentum.

The shapes of the distributions, however, differ considerably. Indeed, at the RESI threshold intensity [Fig. 2(a)], one observes ring-shaped distributions. As the intensity increases, this distribution becomes more and more elongated along the $p_{n\parallel}$ axis [Fig. 2(b)], until the maxima merge and cross-shaped distributions are observed [Fig. 2(c)].

This change of shape may be understood by analyzing the momentum-space constraints. The widths of the distributions are determined by the tunnel ionization of the second electron from an excited state. This process has no classical counterpart and leads to distributions peaked at $p_{2\parallel} = 0$ and which vanish at $p_{2\parallel} = \pm 2\sqrt{U_p}$ (i.e., at the direct ATI cutoff). Increasing the intensity only makes the effective potential barrier smaller or wider, and thus affects the overall yield, but does not change such constraints.

The elongations in the distributions are determined by the rescattering of the first electron. This rescattering, in contrast, delimits a momentum region which is highly dependent on the driving-field intensity. Therefore, its width in momentum space varies. Specifically, at the RESI threshold, there are maxima in the distributions at $p_{1\parallel} = \pm 2\sqrt{U_p}$ due to the fact that the first electron rescatters most probably at a field crossing. However, as these are the only classically allowed

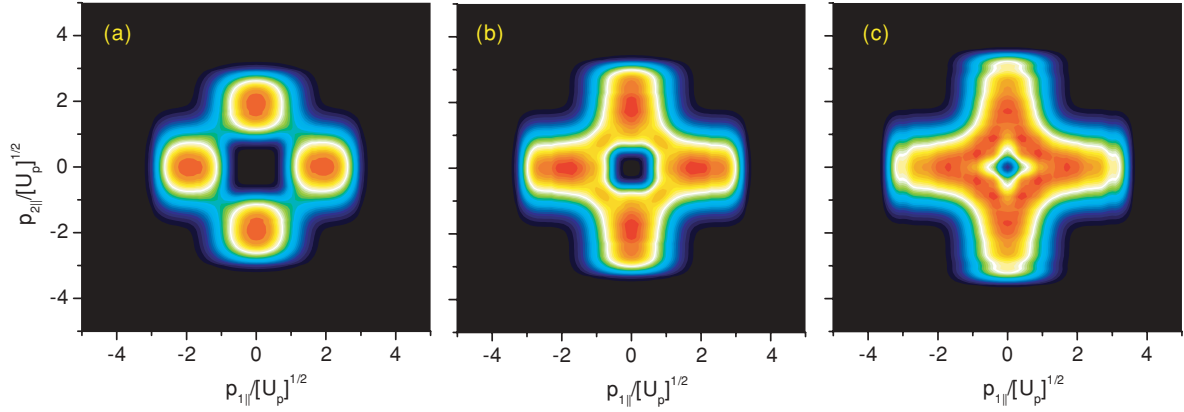


FIG. 2. (Color online) Electron momentum distributions for helium ($E_{1g} = 0.97$ a.u., $E_{2g} = 2$ a.u., and $E_{2e} = 0.5$ a.u.) in a linearly polarized, monochromatic field of frequency $\omega = 0.057$ a.u. All prefactors are considered to be constant. Driving-field intensities are (a) $I = 2.16 \times 10^{14}$ W/cm², (b) $I = 2.5 \times 10^{14}$ W/cm², and (c) $I = 3 \times 10^{14}$ W/cm², respectively. The contour plots have been normalized to the maximum probability in each panel.

momenta, the distributions are fairly narrow around this value. With increasing driving-field intensity, the classically allowed region defined by Eq. (15) becomes more and more extensive and this causes the elongation.

Note that the electrons are indistinguishable, so the above-stated arguments hold upon the exchange $p_{1\parallel} \leftrightarrow p_{2\parallel}$. Hence, the horizontal and vertical axes in the parallel momentum plane are equally affected.

B. Bound-state signatures

We now investigate how the shape of the bound state to which the second electron is excited is imprinted on the electron momentum distributions. We also employ different gauges and types of electron-electron interaction. Explicitly, we assume that the second electron is either excited by a contact-type interaction $V_{12}^{(\delta)}(\mathbf{r}_1 - \mathbf{r}_2) = \delta(\mathbf{r}_1 - \mathbf{r}_2)$ or by a long-range, Coulomb-type interaction $V_{12}^{(C)}(\mathbf{r}_1 - \mathbf{r}_2) = 1/|\mathbf{r}_1 - \mathbf{r}_2|$. In the former case, $V_{12}^{(\delta)}(\mathbf{p}_1 - \mathbf{k}) = \text{const}$, while in the latter case $V_{12}^{(C)}(\mathbf{p}_1 - \mathbf{k}) \sim 1/(\mathbf{p}_1 - \mathbf{k})^2$. We are considering these two types of interaction in order to make an assessment of the role of the Coulomb tail in the RESI mechanism. In electron-impact ionization, for instance, the long-range nature of the Coulomb interaction leads to a V-shaped structure [7,11,12].

In order to perform a direct comparison, we take the same parameters as in Fig. 2 but incorporate the prefactors $V_{\mathbf{p}_1 e, \mathbf{k} g}^{(1s \rightarrow 2s)}$ and $V_{\mathbf{p}_2 e}^{(2s)}$, or $V_{\mathbf{p}_1 e, \mathbf{k} g}^{(1s \rightarrow 2p)}$ and $V_{\mathbf{p}_2 e}^{(2p)}$, corresponding to the $1s \rightarrow 2s$ or $1s \rightarrow 2p$ excitation with subsequent tunneling, respectively.

In Fig. 3, we consider the lowest intensity in the previous figure and the velocity gauge. If the electron is excited to the $2s$ state [Fig. 3(a)], we observe four spots which are slightly elongated along the $p_{n\parallel}$ axis. Hence, in comparison to its constant-prefactor counterpart, [i.e., Fig. 2(a)], there was a narrowing. This narrowing is caused by the interplay of two features in the prefactor $V_{\mathbf{p}_2 e}^{(2s)}$. First, this prefactor exhibits two symmetric nodes, which, for vanishing transverse momentum, are located at $p_{2\parallel} = \pm \sqrt{U_p}$. As the transverse momentum increases, these minima move toward vanishing parallel momenta. Second, $V_{\mathbf{p}_2 e}^{(2s)}$ decreases very steeply with

transverse momenta $\mathbf{p}_{2\perp}$. Hence, upon integration over this parameter, the main contributions are caused by small values of $\mathbf{p}_{2\perp}$ and vanish near $p_{2\parallel} = \pm \sqrt{U_p}$.

If, on the other hand, one assumes that the second electron is excited to $2p$, there is both a broadening in the distributions and a splitting in their peaks. These features are depicted in Fig. 3(b). The splitting occurs at the axis $p_{n\parallel} = 0$ ($n = 1, 2$) and is caused by the fact that $V_{\mathbf{p}_2 e}$ exhibits a very pronounced node at vanishing momenta (i.e., exactly where one expects $\text{Im}[t]$ to be minimum and the yield to be maximum). This has been verified by a direct inspection of the radial dependence of Eq. (30) and omitting the $V_{\mathbf{p}_2 e}$ prefactor in our computations.

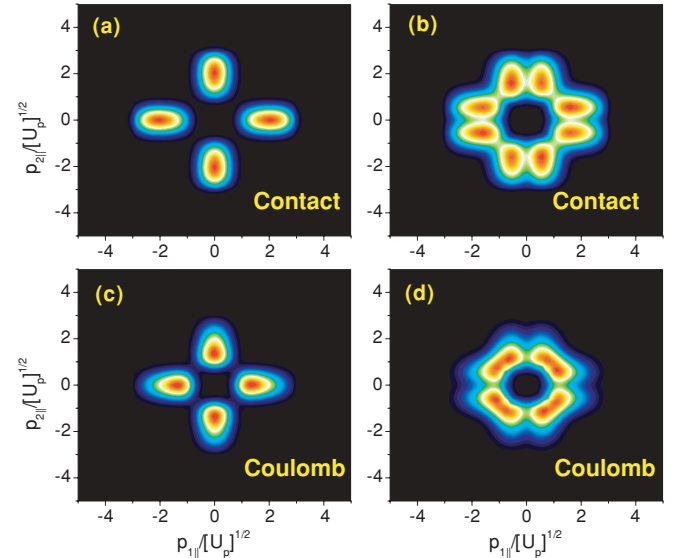


FIG. 3. (Color online) Velocity-gauge electron momentum distributions for helium ($E_{1g} = 0.97$ a.u., $E_{2g} = 2$ a.u., and $E_{2e} = 0.5$ a.u.) in a linearly polarized, monochromatic field of frequency $\omega = 0.057$ a.u. and intensity $I = 2.16 \times 10^{14}$ W/cm². In panels (a) and (c), the first electron has been excited to $2s$, while in panels (b) and (d), it has been excited to $2p$. The interaction employed is indicated in the figure. The contour plots have been normalized to the maximum probability in each panel.

The latter procedure caused the additional minima to disappear (not shown). The broadening in the distributions as compared to the $1s \rightarrow 2s$ case is a consequence of the much slower decrease in $V_{\mathbf{p}_{2e}}^{(2p)}$ with increasing transverse momentum $p_{2\perp}$ and of the absence of the nodes at $p_{2\parallel} = \pm\sqrt{U_p}$. There are also additional nodes at the diagonal $p_{1\parallel} = p_{2\parallel}$ and at the antidiagonal $p_{1\parallel} = -p_{2\parallel}$ of the $p_{1\parallel}p_{2\parallel}$ plane.

In this intensity regime, there seems to be little difference in the shapes of the distributions if the electron is excited to the $2s$ state, regardless of whether the first electron interacts with its parent ion through a contact or a Coulomb interaction [Figs. 3(a) and 3(c), respectively]. This is possibly caused by the fact that the prefactor $V_{\mathbf{p}_{2e}}^{(2s)}$, due to its fast-decaying behavior, delimits a very narrow region in momentum space. This adds up to the very restrictive momentum constraints. In contrast, the effect of the Coulomb tail is much more critical if the electron is promoted to the $2p$ state. Indeed, for a Coulomb-type interaction [Fig. 3(d)], the splitting of the peaks at the axis $p_{n\parallel} = 0$ remains, but the nodes at $p_{1\parallel} = p_{2\parallel}$ and $p_{1\parallel} = -p_{2\parallel}$ disappear as compared to its contact-interaction counterpart [Fig. 3(b)]. This is caused by the fact that the former minima are a characteristic of the $V_{\mathbf{p}_{2e}}^{(2p)}$ prefactor, whereas the latter are mainly determined by momentum-space effects. The Coulomb interaction introduces a further momentum bias and washes out the latter nodes.

We now discuss what happens if the intensity of the driving field is such that $3.17U_p > E_{2g} - E_{2e}$. These results are displayed in Fig. 4, for the highest intensity in Fig. 2. As expected, all distributions are much more elongated along the axis $p_{n\parallel} = 0$, as compared to the low-intensity case. The imprint, however, of the different bound states to which the second electron is excited and from which it subsequently

tunnels are the same as in the below-threshold regime. Indeed, we notice that there is a narrowing in the distributions for the $1s \rightarrow 2s$ case [Figs. 4(a) and 4(c)] and a splitting in the peaks at the four axes in the $1s \rightarrow 2p$ case [Figs. 4(b) and 4(d)]. This is not surprising, as the prefactors $V_{\mathbf{p}_{2e}}^{(2s)}$ and $V_{\mathbf{p}_{2e}}^{(2p)}$ exhibit the same functional dependencies as before.

The shapes of distributions, however, change much more critically in this intensity regime, with regard to the type of electron-electron interaction, than for the intensity used in Fig. 3. For all cases, the distributions computed using the Coulomb-type interaction [Figs. 4(c) and 4(d)] are much more localized in the low-momentum regions than those computed with a contact-type interaction [see Figs. 4(a) and 4(b)]. This is expected, as $V_{12}(\mathbf{p}_1 - \mathbf{k})$ favors low momenta for the former while it is constant for the latter. Physically, this reflects the fact that rescattering of the first electron is now allowed to occur over an extensive region in momentum space. Hence, it does make a difference whether the second electron is excited by a long- or zero-range interaction.

We now perform an analysis of the electron momentum distributions in the length gauge. In this case, the prefactor $V_{\mathbf{p}_{2e}}$ governing the tunneling of the second electron exhibits a singularity and must be incorporated in the action. The modifications in the action read, for the $2s$ and $2p$ bound states,

$$-i\partial_t \ln V_{\mathbf{p}_{2e}}^{(2s)} = -i \frac{2E(t)\tilde{p}_2(t)\{[\tilde{p}_2(t)]^2 + \mathbf{p}_{2\perp}^2 - 6E_{2e}\}}{\chi_+(\tilde{\mathbf{p}}_2(t))\chi_-(\tilde{\mathbf{p}}_2(t))} \quad (40)$$

and

$$-i\partial_t \ln \tilde{V}_{\mathbf{p}_{2e}}^{(2p)} = i \frac{E(t)\{\mathbf{p}_{2\perp}^2 + 2E_{2e} - 3[\tilde{p}_2(t)]^2\}}{\tilde{p}_2(t)\{[\tilde{p}_2(t)]^2 + \mathbf{p}_{2\perp}^2 + 2E_{2e}\}}, \quad (41)$$

respectively, with $\chi_{\pm}(\tilde{\mathbf{p}}_2(t)) = \{[\tilde{p}_2(t)]^2 + \mathbf{p}_{2\perp}^2 \pm 2E_{2e}\}$ and $E(t) = -\partial_t A(t)$.

For each orbit, the tunneling time of the second electron splits into two values, as compared to the nonmodified action. This has particularly important consequences as far as $\text{Im}[t]$ is concerned, since it provides a rough measure of the width of the barrier through which the second electron tunnels. Physically, this means there will be one set of orbits for which the effective potential barrier will be widened, and another one for which it will be narrowed.

In Fig. 5, we present the contributions of each of the orbits resulting from this splitting for a final $2s$ state, for two different driving-field intensities. For simplicity, in order to single out the effect of the modified action, we took the rescattering prefactor $V_{\mathbf{p}_{1e},\mathbf{k}g}$ to be constant.

In general, the distributions differ quantitatively by a factor between 1.5 and 1.7, depending on whether $\text{Im}[t]$ decreased [Figs. 5(a) and 5(c)] or increased [Figs. 5(b) and 5(d)]. This shows that the splitting in this quantity is small and, therefore, both contributions are comparable.

Furthermore, the distributions displayed in Fig. 5 are strikingly similar to those observed in Fig. 2 [see Figs. 2(a) and 2(c), and Figs. 5(a) and 5(c)], for which only constant prefactors have been considered. Indeed, the width of all distributions, along the axis, is determined by the direct ATI cutoff; that is, $-2\sqrt{U_p} \leq p_{n\parallel} \leq 2\sqrt{U_p}$. At first sight, this is unexpected, as we are assuming that the second electron

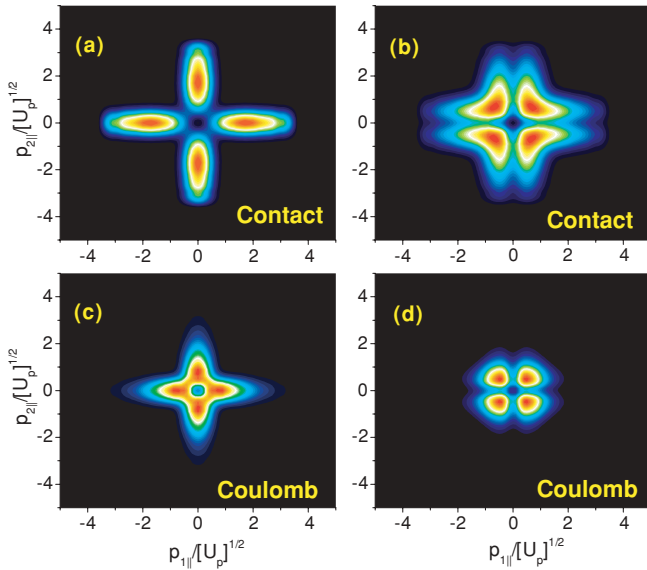


FIG. 4. (Color online) Velocity-gauge electron momentum distributions for the same parameters as in the previous figure, but for driving-field intensity $I = 3 \times 10^{14}$ W/cm². In panels (a) and (c), the first electron has been excited to $2s$, while in panels (b) and (d), it has been excited to $2p$. The interaction employed is indicated in the figure. The contour plots have been normalized to the maximum probability in each panel.

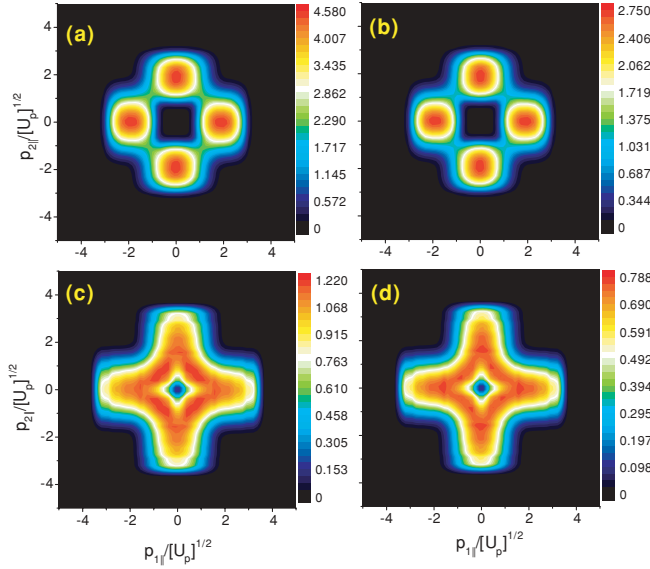


FIG. 5. (Color online) Length-gauge electron momentum distributions for helium in a linearly polarized, monochromatic field. Throughout, we assumed $V_{p_1 e, kg} = \text{const}$, V_{12} is a contact-type interaction, and we incorporated $V_{p_2 e}^{(2s)}$ in the action. Panels (a) and (c) correspond to the trajectories for which the barrier has been narrowed by the modifications in the action, while panels (b) and (d) correspond to those for which it has been widened. The upper and lower panels correspond to driving field intensities $I = 2.16 \times 10^{14} \text{ W/cm}^2$ and $I = 3 \times 10^{14} \text{ W/cm}^2$, respectively. In order to perform a quantitative comparison, we provide the explicit values for the NSDI yield. In the upper and lower panels, these values are multiplied by 10^{17} and 10^{16} , respectively.

is tunneling from a $2s$ state. As previously discussed, the prefactor $V_{p_2 e}^{(2s)}$ exhibits a node in $p_{2||} = \pm \sqrt{U_p}$, which leads to a narrowing of the distributions along the $p_{n||}$ axis. An inspection of Eq. (24) also suggests that, were it not for its singularity, the length-gauge prefactor would be very similar to the velocity-gauge prefactor. This is a consequence of the fact that the second electron is leaving when the field $E(t)$ is near its maximum. For a monochromatic field, this implies that the vector potential $A(t)$ is practically vanishing.

One should note, however, that we are considering only the individual contributions from each of the orbits originating from the modification of the action. It is very likely that, in order to recover the structure determined by the prefactor $V_{p_2 e}^{(2s)}$, one must consider the coherent superposition of all the orbits originating from the splitting of $\text{Im}[t]$ when computing the yield. Since these contributions are comparable, one expects the above-mentioned nodes to be recovered due to quantum-interference effects.

In order to clarify this issue further, we consider a test case in which Gaussian bound states

$$\psi^{(\alpha)}(\mathbf{r}_\alpha) = \left(\frac{E_\alpha}{\pi}\right)^{3/2} \exp(-E_\alpha r_\alpha^2), \quad (42)$$

where $E_\alpha = E_{1g}, E_{2g}$, and E_{2e} , were taken in order to compute the prefactors V_{kg} , $V_{p_2 e}$, and $V_{p_1 e, kg}$. Due to the choice of wave function, the prefactors $V_{p_2 e}$ and V_{kg} are no longer singular in the length gauge.

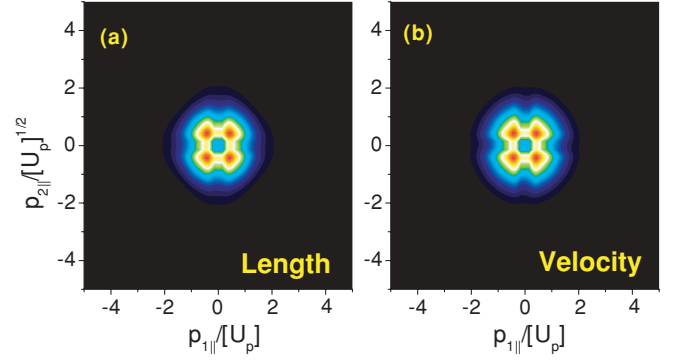


FIG. 6. (Color online) Electron momentum distributions computed with Gaussian bound-state wave functions for the first and second electron, in the (a) velocity and (b) length gauges. Throughout, we considered that the second electron is excited by a Coulomb-type interaction, the same bound-state energies as for helium, and a driving-field intensity $I = 3 \times 10^{14} \text{ W/cm}^2$. The contour plots have been normalized to the maximum probability in each panel.

The explicit expression for the ionization prefactors reads

$$V_{p\zeta} = 2 \left(\frac{E_\alpha}{\tilde{p}} \right) \exp\left(-\frac{\tilde{p}^2}{4E_\alpha}\right) \text{erfi}\left(\frac{\tilde{p}}{2\sqrt{E_\alpha}}\right), \quad (43)$$

where erfi indicates the imaginary error function $\text{erfi}(z) = \text{erf}(iz)/iz$, $\tilde{p} = k(t'')$ or $p_2(t)$, $\zeta = g$ or e , and $E_\alpha = E_{1g}$ or E_{2e} . In the length gauge, $p_2(t) = p_2 + A(t)$ and $k(t'') = k + A(t'')$, while, in the velocity gauge, $p_2(t) = p_2$ and $k(t'') = k$. The rescattering prefactor $V_{p_1 e, kg}$ is given by

$$V_{p_1 e, kg} = \left[\frac{E_{2e} E_{2g}}{\pi(E_{2e} + E_{2g})} \right]^{3/2} \exp\left[-\frac{(\mathbf{p}_1 - \mathbf{k})^2}{4(E_{2e} + E_{2g})}\right] \times V_{12}(\mathbf{p}_1 - \mathbf{k}). \quad (44)$$

This prefactor is the same for both velocity and length gauges.

Figure 6 exhibits the results obtained by employing the prefactors (43) and (44) in the velocity and length gauges [Figs. 6(a) and 6(b), respectively]. The figure shows nearly identical electron momentum distributions. This is in agreement with the previous discussion.

C. Comparison with experiments

We now perform a direct comparison with the results in Ref. [15]. In particular, in this reference, the distributions encountered have been modeled by employing the electron-impact ionization physical mechanism and a modified ionization threshold for the second electron. Apart from that, however, in view of the driving-field intensities involved, one expects recollision-excitation tunneling to be present.

For that purpose, we consider argon and the same laser-field parameters as in Ref. [15] (cf. Fig. 2 therein). We assume, however, that, when the first electron recollides, it excites the second electron from the $3p$ state to either the $4s$ or the $4p$ state. Thereby, we took the velocity gauge and assumed that the first electron interacts with the ion by a Coulomb or contact interaction.

The results for the $3p \rightarrow 4s$ excitation are presented in Fig. 7. An overall feature in the distributions are two main maxima along the $p_{n||}$ ($n = 1, 2$) axis. These features are

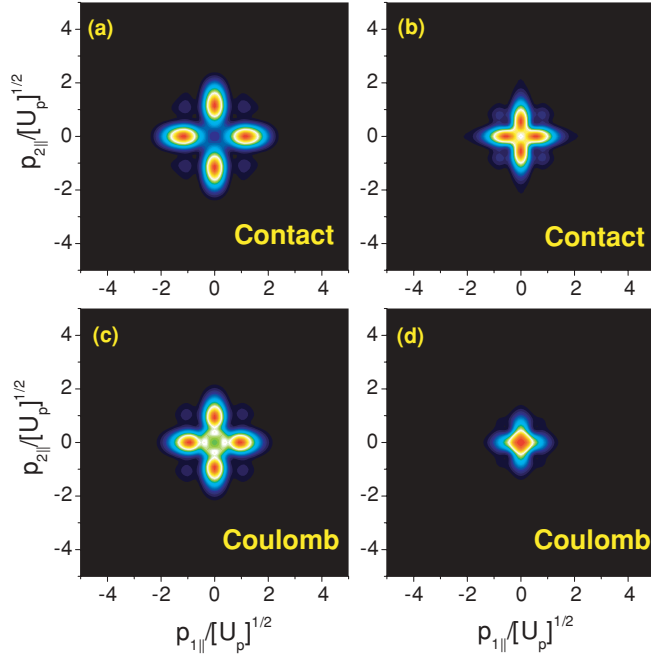


FIG. 7. (Color online) Velocity-gauge electron momentum distributions for argon in a linearly polarized, monochromatic field of frequency $\omega = 0.057$ a.u. The electron is excited from $3p$ to $4s$ (i.e., $E_{1g} = 0.58$ a.u., $E_{2g} = 1.02$ a.u., and $E_{2e} = 0.40$ a.u. in our calculations). The laser-field intensities are (a), (c) $I = 9 \times 10^{13}$ W/cm² and (b), (d) $I = 1.5 \times 10^{14}$ W/cm². The type of interaction taken, V_{12} , is indicated in the figure. The contour plots have been normalized to the maximum probability in each panel. We have verified, however, that the highest yields on left-hand panels are between one and a half and two orders of magnitude smaller than those on the right-hand side.

mainly caused by the $V_{p_{2e}}^{(4s)}$ prefactor for the tunnel ionization of the second electron, which decays very rapidly with increasing transverse momenta and exhibit nodes near $p_{2||} = \pm 0.5\sqrt{U_p}$. In general, we have verified that this prefactor determines the shape of the electron momentum distributions. Secondary maxima, around one order of magnitude smaller, occur due to the rescattering prefactor $V_{p_{1e}, kg}^{(3p \rightarrow 4s)}$. This prefactor exhibits an annular shape around $p_{1||} = p_{2||} = 0$.

The existing experiments, however, do not lead to distributions concentrated along the axis of the $p_{1||}p_{2||}$ plane. The results for helium in the previous section suggest that a p state may lead to broader distributions. For that reason, we assume that, instead, the second electron is excited to the $4p$ state.

Figure 8 depicts the electron momentum distributions for argon under the assumption that the electron was excited from $3p$ to $4p$. All distributions in the figure exhibit four main maxima, which are broader than those in Fig. 7 and almost split at the axis $p_{n||} = 0$. These maxima are mainly determined by the prefactor $V_{p_{2e}}^{(4p)}$, which has a node at the axis for low transverse momenta and nodes around $p_{2||} = \pm\sqrt{U_p}$ across a wide transverse-momentum range. Apart from that, the prefactors $V_{p_{1e}, kg}^{(3p \rightarrow 4p)}$ decay more slowly with regard to the transverse momenta. This implies that, upon integration, a larger momentum region is contributing to the NSDI yields. As in the previous case, this prefactor also leads to secondary

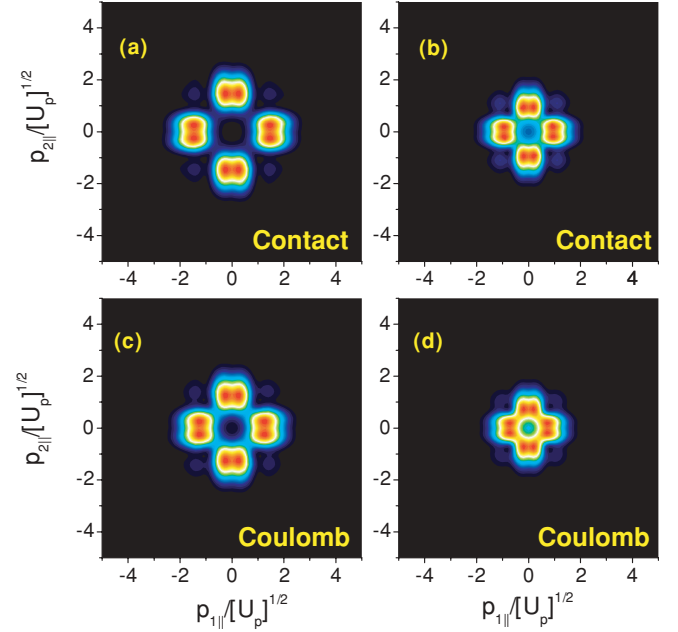


FIG. 8. (Color online) Velocity-gauge electron momentum distributions for aArgon in a linearly polarized, monochromatic field of frequency $\omega = 0.057$ a.u. The electron is excited from $3p$ to $4p$ (i.e., $E_{1g} = 0.58$ a.u., $E_{2g} = 1.02$ a.u., and $E_{2e} = 0.31$ a.u. in our calculations). The laser-field intensity is (a), (c) $I = 9 \times 10^{13}$ W/cm² and (b), (d) $I = 1.5 \times 10^{14}$ W/cm². The type of interaction V_{12} taken is indicated in the figure. The contour plots have been normalized to the maximum probability in each panel. We have verified, however, that the highest yields on left-hand panels are between one and two orders of magnitude smaller than those on the right-hand side.

maxima [see Figs. 8(c) and 8(d) for concrete examples]. In all cases, in both Figs. 7 and 8, a Coulomb-type interaction mainly introduces a bias toward lower momenta.

Despite the above-mentioned broadening, the electron momentum distributions in Fig. 8 are still considerably narrower than those observed in Ref. [15]. Within our framework, this constraint is imposed by the $V_{p_{2e}}^{(4p)}$ prefactor. In fact, we have verified that, for large principal quantum number, this prefactor always exhibits nodes at lower absolute momenta than the ATI cutoff of $p_{2||} = 2\sqrt{U_p}$. In fact, if $V_{p_{2e}}$ is taken to be constant, the distributions become considerably broader and a better agreement with the experiments is obtained. This is shown in Fig. 9 as ring-shaped distributions with four symmetric maxima at $p_{1||} = p_{2||}$ and $p_{1||} = -p_{2||}$. Such maxima are mainly determined by the $V_{p_{1e}, kg}^{(3p \rightarrow 4p)}$ prefactor. One should note, however, that this procedure is inconsistent from a theoretical perspective: Since the electron has been excited to the $4p$ state, it should subsequently tunnel from it. Hence, the pertaining prefactor must be taken.

Nonetheless, the procedure of removing the $V_{p_{2e}}$ prefactor appears to mimic to some extent what happens in a real-life situation, for which the bound states exhibit ac Stark shifts. The first electron returns near a field crossing. Hence, at the rescattering time t' , these shifts are only a small perturbation and are not expected to influence the distributions significantly. The second electron, however, tunnels close to a field maximum. This means that, in this case, the Stark

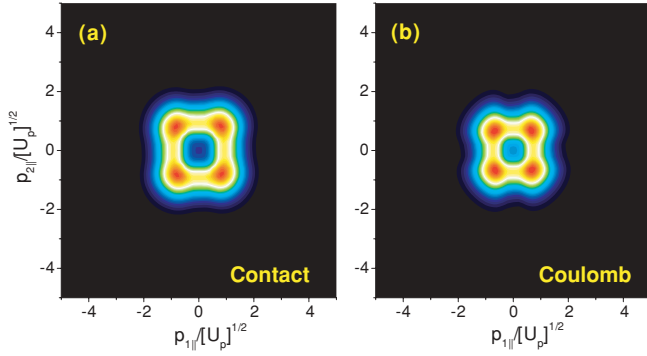


FIG. 9. (Color online) Velocity-gauge electron momentum distributions for argon in a linearly polarized, monochromatic field of frequency $\omega = 0.057$ a.u. and intensity $I = 1.5 \times 10^{14}$ W/cm². The electron is excited from $3p$ to $4p$. We have taken the prefactor $V_{p2,e}$ to be constant. The type of interaction V_{12} is indicated in the figure. The contour plots have been normalized to the maximum probability in each panel.

shifts will be more prominent and will cause a broadening in the electron momentum distributions. A rough, ad hoc way of verifying this is to introduce a time-dependent excited-state energy according to the procedure described in Ref. [30]. Such an approach was also adopted in Ref. [15] for electron-impact ionization. A rigorous way of incorporating these shifts, however, is far more challenging.

IV. CONCLUSIONS

Our analysis of the RESI mechanism shows that the NSDI electron momentum distributions depend on the interplay between the relevant momentum-space regions, the type of interaction exciting the second electron, and the spatial dependence of the bound states involved. We commence by discussing each of these issues separately.

The shapes of the electron momentum distributions are determined by the interplay between two different behaviors associated with the collision of the first electron and the tunneling of the second electron. The momentum region determined by the tunnel ionization of the second electron from an excited state will always be restricted by the direct ATI cutoff. The relevant momentum region does not change regardless of the driving-field intensity, as this will always be a classically forbidden process.

The first electron, on the other hand, rescatters inelastically with its parent ion, giving part of its kinetic energy upon return to excite the second electron. Hence, if its maximum return energy is larger than the energy difference $E_{2g} - E_{2e}$, rescattering has a classical counterpart. This implies that there will be a classically allowed region in momentum space. If, however, this energy is just enough to excite the second electron, the classical region collapses. Hence, the extension of the relevant region in momentum space related to the rescattering of the first electron depends on the driving-field intensity. Hence, the distributions become increasingly elongated as the intensity increases.

This also implies that one may define a threshold driving-field intensity for the RESI mechanism. This intensity is considerably lower than that necessary for the second ionization

potential to be overcome by the second electron (i.e., for electron-impact ionization to occur).

Apart from that, we have observed that the bound states involved in the process leave very distinct fingerprints on the electron momentum distributions. This is particularly true for the bound state of the second electron, prior and subsequently to excitation. In fact, the widths of the distributions, their shapes, and the number of maxima present strongly depend on the principal and orbital quantum numbers of the bound states involved.

In contrast, the type of interaction V_{12} by which the second electron is excited influences such distributions in a less drastic way. Indeed, a long-range, Coulomb interaction mainly introduces a bias toward lower momenta, as compared to a contact-type interaction.

A very important observation is that all distributions encountered in this work are equally spread over the four quadrants of the $p_{1||}p_{2||}$ plane. Under no circumstances have we found electron momentum distributions concentrated only on the second and fourth quadrants of this plane, as reported in the literature [18,20,21].

Within our framework, the above-stated symmetry can immediately be inferred from Eq. (39). Nonetheless, one could argue that our approach does not include the residual binding potential in the electron propagation in the continuum. Recent results, however, from a classical-trajectory computation in which the Coulomb potential has been incorporated, also revealed the same symmetry if only the RESI mechanism is singled out [25] (see Figs. 3(b) and 4(b) in Ref. [25]). This is a strong hint that our results are not an artifact of the strong-field approximation.

Hence, we suspect that, in the existing literature, the contributions from the RESI mechanism to nonsequential double ionization also equally occupy the four quadrants of the $p_{1||}p_{2||}$ plane. They may, however, be difficult to extract, as explained below.

In many situations addressed in the literature, the driving-field intensity is high enough for electron-impact ionization to occur. This means that this latter NSDI mechanism is also present, and it fills the first and third quadrants of the $p_{1||}p_{2||}$ plane. Since, in many *ab initio* models, the different rescattering mechanisms are difficult to disentangle, the contributions from electron-impact ionization possibly obscure those from RESI in this region. In the second and fourth quadrant of the parallel momentum plane, the former contributions are absent and those from RESI can be more easily identified. In our approach, electron-impact ionization is absent from the start. Hence, we can study the RESI process separately and assess all its consequences.

Furthermore, in several classical computations, one tries to mimic the RESI mechanism by considering, instead, electron impact ionization with a time delay between the rescattering of the first electron and the rescattering of the second electron. This is a completely different physical process, which populates only the second and fourth quadrants of the parallel momentum plane [18,20,21,24].

In general, the RESI mechanism becomes more prominent as the driving-field intensity is far below the ionization threshold, so the second electron may no longer be provided with enough energy to overcome the second ionization

potential. Alternatively, if the pulse is long, one also expects RESI to be more prominent [17]. Otherwise, electron-impact ionization will dominate.

An example of the former scenario can be found in Ref. [15] (see Fig. 2 therein) for NSDI around the electron-impact ionization threshold. For the higher driving-field intensity, one may identify two elongated shapes along $p_{1\parallel} = p_{2\parallel}$ superposed to a more symmetric structure. These shapes are a signature of the electron-impact ionization mechanism. As the intensity decreases, this process becomes less and less prominent and RESI dominates. Consequently, the distributions become symmetric and equally occupy the four quadrants of the parallel momentum plane. In fact, ring-shaped distributions centered around $p_{1\parallel} = p_{2\parallel} = 0$ have been observed experimentally for this intensity region [15,16].

Our results are far more localized near the $p_{n\parallel} = 0$ axis than the experimental findings. This discrepancy may be due to the following reasons. First, for the higher intensities employed in Ref. [15], collisional excitation may take place not only to the $4s$ or to the $4p$ state but also to highly lying states, or to a coherent superposition of excited states. This may have to be taken into account in order to reproduce the experimental data.

Second, at the relevant driving-field intensities, one expects the excited states to be distorted by the field, and the propagation of the electron in the continuum near the core to be influenced by the residual ionic potential. This implies that a semianalytical treatment beyond the strong-field approximation is necessary (see, e.g., [13,31] for other phenomena and the electron-impact ionization case, respectively). The Stark shift of the excited bound state seems to be particularly important, due to the fact that the second electron leaves close to a field maximum. One should note, however, that a rigorous treatment of such features is very demanding and is not performed in this work.

Finally, the results obtained in this paper show that the dynamics of the second electron have a huge influence on the shapes of the electron momentum distributions. A suppression or enhancement in its tunnel ionization would lead to an overall suppression or enhancement in the yield. Furthermore, changes in its ionization time would affect its most probable momentum and, therefore, the peaks of the electron momentum distributions. Apart from that, depending on the shape of the excited-state wave function, there may be nodes in the corresponding prefactors which may cause the distributions to broaden or narrow. This means that, in principle, the shapes of the electron momentum distributions can be manipulated by an adequate driving-field choice and by preparing the first electron in a suitable superposition of states.

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APPENDIX A

In this Appendix, we provide the general expressions for the prefactors employed in this paper related to hydrogenic states. We make no simplifying assumption on the initial states of the first and second electron, and on the excited state to which the second electron is promoted, apart from the fact that they are given by hydrogenic wave functions. In order to compute the prefactors, we employ the expansion

$$e^{-i\mathbf{q}\cdot\mathbf{r}_\alpha} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l j_l(qr_\alpha) Y_l^m(\theta_{q_\alpha}, \varphi_{q_\alpha}) [Y_l^{m'}(\theta_\alpha, \varphi_\alpha)]^*, \quad (\text{A1})$$

where $\mathbf{q}$ denotes a generic momentum, $\mathbf{r}_\alpha$ the coordinate of the α th electron, and $j_l(\cdot)$ the spherical Bessel functions of the first kind. This expression is used in the derivation of both $V_{\mathbf{p}_{1e}, \mathbf{k}g}$, together with the orthogonality relation

$$\int [Y_l^{m'}(\theta_\alpha, \varphi_\alpha)]^* Y_l^m(\theta_\alpha, \varphi_\alpha) d\Omega = \delta_{ll'} \delta_{mm'}, \quad (\text{A2})$$

where Ω denotes the solid angle.

For the former prefactor, Eq. (5) reduces to

$$\begin{aligned} V_{\mathbf{p}_{2e}} &\sim \iint \exp\{-i[\tilde{\mathbf{p}}_2(t) \cdot \mathbf{r}_2]\} R_{nl}(r_2) Y_l^m(\theta_2, \varphi_2) r_2 dr_2 d\Omega \\ &= 4\pi (-i)^l Y_l^m(\theta_{\tilde{\mathbf{p}}_2}, \varphi_{\tilde{\mathbf{p}}_2}) \mathcal{I}_1, \end{aligned} \quad (\text{A3})$$

with

$$\mathcal{I}_1 = \int_0^\infty r_2 R_{nl}(r_2) j_l[\tilde{p}_2(t)r_2] dr_2. \quad (\text{A4})$$

Similarly, Eq. (6) reads

$$V_{\mathbf{p}_{1e}, \mathbf{k}g} \sim V_{12}(\mathbf{p}_1 - \mathbf{k}) \mathcal{I}_2, \quad (\text{A5})$$

with

$$\begin{aligned} \mathcal{I}_2 &= \int d^3r_2 e^{i(\mathbf{k}-\mathbf{p}_1)\cdot\mathbf{r}_2} R_{n_e l_e}(r_2) [Y_{l_e}^{m_e}(\theta_2, \varphi_2)]^* \\ &\quad \times R_{n_g l_g}(r_2) Y_{l_g}^{m_g}(\theta_2, \varphi_2), \end{aligned} \quad (\text{A6})$$

where the indices g and e in the principal, orbital, and magnetic quantum numbers refer to the ground and excited states, respectively.

We now compute the radial integrals $\mathcal{I}_1$ and $\mathcal{I}_2$ explicitly. For that purpose, let us consider a generic hydrogenic radial wave function

$$\begin{aligned} R_{nl}(r) &= C_{nl} r^l \exp[-\sqrt{2E_n}r] \\ &\quad \times \sum_{v=0}^{n-l-1} \frac{2^v (-1)^{v+1} (\sqrt{2E_n})^v}{v!(n-l-1-v)!(2l+1+v)!}, \end{aligned} \quad (\text{A7})$$

with

$$C_{nl} = - \left\{ \frac{(2\sqrt{2E_n})^{3+2l} (n-l-1)!}{2n[(n+l)^3]} \right\}^{1/2} [(n+l)^2]. \quad (\text{A8})$$

In the above-stated equations, E_n denotes the energy of the bound state to be studied, that is, $n = 2g$ or $n = 2e$ for the

ground or excited states of the second electron, respectively. Since we are performing a qualitative analysis, we concentrate mostly on the functional form of $R_{nl}(r)$. The integral $\mathcal{I}_1$ present in the prefactor $V_{\mathbf{p}_{2e}}$ can then be written as

$$\mathcal{I}_1 \propto \sum_{\nu=0}^{n-l-1} \frac{(-1)^{\nu+1} 2^{\nu-1-l} (\sqrt{2E_n})^{-2-l}}{\nu!(n-l-1-\nu)!(2l+1+\nu)!} \frac{\Gamma(2+\nu+2l)}{\Gamma(3/2+l)} \\ \times {}_2F_1\left(1+l+\frac{\nu}{2}, \frac{3+\nu}{2}+l, \frac{3}{2}+l, -\frac{[\tilde{p}_2(t)]^2}{2E_e}\right), \quad (\text{A9})$$

where ${}_2F_1(\cdot)$ denote hypergeometric functions.

The integral $\mathcal{I}_2$ in $V_{\mathbf{p}_{1e}, \mathbf{k}_g}$ is slightly more involved. It may be explicitly written as

$$\mathcal{I}_2 = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l Y_l^m(\theta_{q_a}, \varphi_{q_a}) \mathcal{I}_{2R} \mathcal{I}_{2\Omega}, \quad (\text{A10})$$

where

$$\mathcal{I}_{2R} = \int_0^{\infty} r_2^2 R_{n_g l_g}(r_2) R_{n_e l_e}(r_2) j_l(\kappa r_2) dr_2 \quad (\text{A11})$$

and

$$\mathcal{I}_{2\Omega} = \int Y_l^m(\theta_2, \varphi_2) [Y_{l_e}^{m_e}(\theta_2, \varphi_2)]^* Y_{l_g}^{m_g}(\theta_2, \varphi_2) d\Omega \quad (\text{A12})$$

give the radial and angular dependencies of such prefactors, respectively. The explicit expression for $\mathcal{I}_2$ is then

$$\mathcal{I}_2 = 4\pi \sum_{l=|l_g-l_e|}^{l_g+l_e} \sum_{m=-l}^l (-i)^l (-1)^{m_e} Y_l^m(\theta_{\kappa}, \varphi_{\kappa}) \\ \times \sqrt{\frac{(2l_g+1)(2l_e+1)}{4\pi(2l+1)}} \langle l_g, l_e, 0, 0, |l, 0 \rangle \\ \times \langle l_g, l_e, m_g, -m_e, |l, 0 \rangle \mathcal{I}_{2R}. \quad (\text{A13})$$

The radial integral $\mathcal{I}_{2R}$ is proportional to

$$\mathcal{I}_{2R} \propto \sum_{\nu_g=0}^{n_g-l_g-1} \sum_{\nu_e=0}^{n_e-l_e-1} \frac{(-1)^{\nu_e+\nu_g+1} 2^{\nu_e+\nu_g-1-l} (\sqrt{2E_{2g}})^{\nu_g} (\sqrt{2E_{2e}})^{\nu_e} [\zeta(E_{2g}, E_{2e})]^{-3-\nu_e-\nu_g-l_e-l_g}}{\nu_e! \nu_g! (n_e-l_e-1-\nu_e)! (n_g-l_g-1-\nu_g)! (2l_e+1+\nu_e)! (2l_g+1+\nu_g)!} \\ \times \frac{\Gamma[2+\lambda]}{\Gamma[3/2+l]} \left[\frac{\kappa^2}{\zeta^2(E_{2g}, E_{2e})} \right]^{l/2} {}_2F_1\left(\frac{3+\lambda}{2}, \frac{4+\lambda}{2}, \frac{3}{2}+l, -\frac{\kappa^2}{\zeta^2(E_{2g}, E_{2e})}\right), \quad (\text{A14})$$

where $\lambda = \nu_e + \nu_g + l_e + l_g + l$ and $\zeta(E_{2g}, E_{2e})$ is defined according to Eq. (23). Note that the terms in Eq. (A13) are only nonvanishing if $m = m_g - m_e$ and $l_1 + l_2 - l$ is even.

In the present work, apart from the case in which only s states are involved and the angular integrals are constant, one may identify the following cases. First, the second electron may initially be in a p state and be excited to an s state. In this case, $l = l_g = 1$ and $l_e = 0$. Second, if the electron is initially in an s state and is excited to a p state, then $l = l_e = 1$ and $l_g = 0$. Finally, if the second electron suffers a transition from a p state to another p state, in principle $l = 0, 1, 2$. Due to the constraints upon l for the Clebsch-Gordan coefficients, however, only the terms with $l = 0, 2$ survive. Apart from that, the constraint upon m imposes further restrictions for m_e and m_g . The above-stated expressions, however, are applicable to generic hydrogenic states.

APPENDIX B

In this Appendix, we provide an argument for neglecting the interference terms in Eq. (39). Explicitly, we wish to show that

$$\iint |M_R + M_L|^2 d^2 p_{1\perp} d^2 p_{2\perp} \\ \simeq \iint |M_R|^2 + |M_L|^2 d^2 p_{1\perp} d^2 p_{2\perp}. \quad (\text{B1})$$

The integrand in each of these transition amplitudes can be written as $\mathcal{F}_{\xi}(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k}) \exp(iS_{\xi})$, where $\mathcal{F}_{\xi}(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k})$ is the product of all the prefactors involved and ξ refers to the left or right peak. The

corresponding electron orbits are displaced by half a cycle. Since the first and second electrons tunnel near $A(t'') = A(t) = 0$, $\mathcal{F}_R(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k}) \simeq \mathcal{F}_L(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k}) = \mathcal{F}(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k})$. For a linearly polarized monochromatic field, the action related to the orbits with the same momenta $\mathbf{p}_1, \mathbf{p}_2$ is $S_L = \alpha(\mathbf{p}_1, \mathbf{p}_2, t, t') + S_R$. Thereby,

$$\alpha(\mathbf{p}_1, \mathbf{p}_2, t, t') = \alpha_0(p_{1\parallel}, p_{2\parallel}, t, t') + \frac{\mathbf{p}_{1\perp}^2}{2} + \frac{\mathbf{p}_{2\perp}^2}{2},$$

where

$$\alpha_0(p_{1\parallel}, p_{2\parallel}, t, t') = 2U_p + \frac{p_{1\parallel}^2}{2} + \frac{p_{2\parallel}^2}{2} + E_{1g} + E_{2g} \\ - \frac{2A_0}{\omega} [p_{1\parallel} \sin(\omega t') + p_{2\parallel} \sin(\omega t)]. \quad (\text{B2})$$

Hence,

$$|M_R + M_L|^2 \simeq |\exp(iS_R)|^2 \mathcal{F}(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k}) \\ \times \left[2 + 2 \cos \left(\frac{\alpha_0}{2} + \frac{\mathbf{p}_{1\perp}^2}{4} + \frac{\mathbf{p}_{2\perp}^2}{4} \right) \right]. \quad (\text{B3})$$

The first term in Eq. (B3) corresponds to the incoherent sum, while the second term gives the interference condition. If the prefactors are slowly varying, their dependence with regard to the transverse momentum can be neglected to first approximation. Therefore, the dependence with regard to $\mathbf{p}_{n\perp}^2$ ($n = 1, 2$) is mainly determined by the trigonometric function coming from the interference between the actions S_R and S_L . Upon integration over $p_{1\perp}^2$ and $p_{2\perp}^2$,

they cause the interference terms to vanish. In practice, this means that the contributions from the coherent term

are small. If $\mathcal{F}(\mathbf{p}_1, \mathbf{p}_2, t, t', t'', \mathbf{k}) = \text{const}$, this argument is exact.

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